

Internship Report: Data Engineering

Company: Cod<avatar/> tech Pvt. Ltd

Location: Kalopul, Kathmandu Nepal

Period: March 17, 2026 – May 15, 2026

Designation: Data Engineering Intern

1. Introduction and Objective

The primary objective of this internship at Codavatar was to transition the theoretical computer science concept to practical implementation. I aimed to move from being the “tool guy” to moving on be the “builder guy” and so that I’m able to develop some problem-solving skills

Core Goals:

- Understanding of big data Vitesse and volume stacks
- Implement real time data processing pipelines
- Apply object-oriented Programming (OOP) to data infrastructure

2. Learning Experience

During my internship, I learnt many data engineering concepts that helped me understanding, here are the things I learned

Foundational Engineering

- **Linux and Bash:** Developed scripts for environment automation and terminal based system
- **Advanced Python:** Mastered OOP principles, to enhance code for better readability

Database Management

Relational Database: Used SQL Alchemy to bridge the gap between pythonic object-oriented code and relational data storage

NoSQL and Search Engine:

- **MongoDB:** Explored document-oriented storage for flexible and unstructured data
- **Elasticsearch:** Implemented text analysis and custom mapping to handle high speed data retrievals
- **Graph Database (Neo4j):** Specialized in relationship heavy data. Learned some cypher query language to implement a python client to load and query connections that traditional database struggle to map

Big Data Ecosystem

- **Kafka and Streaming:** Developed a deep understanding of real time architecture. Configured Kafka producer and consumer, and integrated Spark streaming for mini-batch processing

Automation and CI/CD

- **Containerization (Docker):** Used Docker to package applications into portable containers, ensuring the flexible to run different application needed to run the project

3. Project Implementation: Real-Time Impact System

a) **The Problem:** Traditional databases are often too slow to map how high-demand, real-time events (a major festival or concert) impact surrounding infrastructure and availability.

b) **Builder Solution:**

- **Stream Ingestion:** Used Kafka to ingest real-time data flows.
- **Distributed Processing:** Processed the stream via PySpark to clean and transform the data on the fly.
- **Relationship Mapping:** Pushed the results into Neo4j to visualize and query the geographical and social "impact" connections between events and properties.
- **Deployment:** The entire pipeline was orchestrated using Docker, ensuring a modular and scalable architecture.

4. Expectations vs. Reality

- **Expectation:** I initially thought data engineering was simply about being a tool guy who know just to use tools such as pandas, NumPy, python, SQL but I was wrong
- **Reality:** I discovered that from transitioning to being a tool guy to being a "builder" requires a deep understanding of Data Storytelling, what outcome our data wants from us and the stakeholders.
- **Achievement:** I successfully transitioned from just knowing how a tool works to knowing when and why to use it to solve a specific problem.

5. Conclusion

In conclusion, my two months at Codavatar have been a period of intense technical growth. This internship enhanced my Data Engineering knowledge, specially working with Big Data which I never had, such as Kafka, Spark and improved my communication, teamwork and problem-solving skills.

This experience gained during the internship will be very helpful for my future career in Data engineering moving forward.